

CHE 205, Fall 2019, Problem Set 2

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Due date: 6:00pm, Fri, 09/13/2019

All problems are from the textbook by V.L. Law.

Submit 2 files on blackboard. For each problem, submit 1 file. Create different tabs (sheets) for different parts (a, b, c, etc.) of the same problem.

Problem 1 (10 pts). Exercise 2.1.

Problem 2 (10 pts). Exercise 2.2.

Exercise 2.1: Further test the VBA program of Example 2.6 as follows:

- a. Put in two initial points that *do not* bracket the roots. This should produce the appropriate error message.

- b. Test the program using different initial points. Some suggested ones are (0.1, 1.5) and (0.01, 1.55).
- c. Test the program for the van der Waals problem as given in Example 1.1 of Chapter 1. Choose two initial points from the graph of the function shown in Figure 1.1. Shown below is one way to alter the Function Subprogram Fcalc for this problem:

```
Function Fcalc (V)
Dim R, T, P, Tc, Pc ,a, b As Double

R = 0.08206
T = 250
P = 10
Tc = 407.5
Pc = 111.3
a = 4.238448175
b = 0.037555537

Fcalc = (P + a / V ^ 2) * (V - b) - R * T

End Function
```

Exercise 2.2: In order to appreciate the increased efficiency of the modified false position method, redesign and reprogram Example 2.6 so that the original false position method is implemented.

- a. Compare the output of the program for the same test function as Example 2.6 and using the same initial points with that of Example 2.6.
- b. Repeat part a using (0.01, 1.55) as the initial point. Does the algorithm converge?
- c. Repeat this exercise using the van der Waals function as described in Exercise 2.1c.